

Reflective Journal week by week

Week 1 (Jan 5 - 9)

- In Week 1, I was just getting started in my new coop, I had been introduced by my supervisor Kierstan to both the other YorkU co-ops and the full time employees that were a part of the same digital team as I was. As one of the new developers part of the digital team, we had a meeting with the full time developers to give us a formal onboarding and introduction, where they had mentioned all of the software and programs they use and that we would need to get from IT. As a part of this coop, we are working on creating educational games and interactives as part of the Ontario Science Center's mission for both applications on the web and through exhibits. This sort of application was something that I had not done before, and was very unfamiliar with game engines like Godot and programs like GdScript. As we were still getting our programs and laptops, I had taken the time to learn more about Godots structure and workspace to then help me get through that initial learning curve when I officially start work with that engine. Besides the team, we were introduced to the other teams that were a part of the same mission, both a science team and a writers team that work alongside our team to create these interactive experiences for children around the age of 12. This first week had taught me alot about how our team works and communicates with the other teams on this project, and what we could expect in the coming weeks.

Week 2 (Jan 12-16)

- In Week 2, we had finally gotten the programs and devices required for our time in the coop placement, all of the coop students on the team had gotten onboarding packages that would be worked on for both this week and the next. From my previous coop experience, I had already been familiar with the onboarding process and this one was no different, it helped that this time I had a team with other coop students that were in the same situation as I was, which helped with getting along with one another. In these initial weeks I had gotten comfortable enough with others on the team to ask questions and for help with the onboarding process, making things easier for me. With consistent morning meetings and everyone being in office, there was plenty of opportunities for communication with one another, something that I had found to be a positive experience that I wish I had more of at my last coop. It was clear to me that everyone was motivated and passionate about their role and position on the team, and are always there to help one another to get the job done.

Week 3 (Jan 19-23)

- By Week 3, we had been assigned our first official tasks within the Godot application, a minor assignment, playtesting and fixing bugs from the games created by the previous coop. This week had a focus on Salmon Run, a mobile game focused on animal migration. I had enjoyed the playtesting, it gave me a sense of what was expected of me for my coop placement. I had a lot of ideas, both for improving the work of the previous coops and ideas of my own that I wanted to bring up for this next wave of games. Joey and I as the coop developers, had both agreed that the game had some issues that needed to be fixed. Both the ui elements and the flow of the game had issues that never made it around to be fixed, particularly with how your energy and stamina was tracked

and with how animals that were in your path had interacted with the salmon. It was great to see that our fresh minds on the project with the various feedback and ideas were not ignored but but considered by others on the team. All 4 coops, 2 Ui/Ux/Game Art designers (Jessica and Sepanta) and 2 developers had alot of experience of playing games, both of the ui coops were also familiar with game development from the courses they had taken in university which helped in knowing as I would later ask them questions that would help me learn more about the process. This marked a great opportunity as a bigger team than last year to take the experiences and ideas we have to then make new game projects that would be better than of last year. We had understood that there was deadline for the end of our coop with these new game projects, whatever they would end up being (at week 3 was not known)

- I had familiarized myself with the work process of general meetings with the science and writing team and our daily scrums with just the digital team to share whats on our plate and to take the time to brainstorm game ideas
- Salmon Run was my first try at a real Godot application which went smoothly because of my learning done in the previous weeks
- I was still trying to get familiar with the software and version control process with git, but Alicia Bhavik and Jon (full time developers) were very helpful in the process
- I had been able to ask them more about the experience with the various programs and at Ontario Science Centre, but they had joined the team last year so they were still getting used to the process as well with godot being a tool that everyone was still learning, so if we had questions we would discuss with one another if someone had figured out or learned something new about the engine that would help everyone with their assignments and tasks.
- By the end of this week I had made the new changes to the Salmon Run, including a new energy bar that better resembled energy, making it more clear to the user playing
- I had also made updates to how the obstacles and enemies visually appeared in the infinite runner, making them feel more dynamic then static objects floating down a river
- I was glad that my suggestions had made a positive difference to this game project and to people who played it
- Since the team previously had a mobile first approach, testing on both mobile and pc had the same screen shape, which I had mentioned made it not organic on a computer screen
- I had to learn the ftp process filezilla that the team uses for testing game web exports, which I had asked questions about and learned how to then upload the updates made to Salmon Run

Week 4 (Jan 26-30)

- In week 4, Joey had taken over as the main developer who is continuing updates to Salmon Run. While I had moved to the other coop game project Solar Escape
- Solar Escape was also in a mobile only format but was both played for pc and mobile, I was curious and asked questions as to why certain aspects were decided for both games, but ultimately it was because they expect users to be accessing the future digital site from their mobile device or something similar, this would later change as I push for a

more dynamic game approach that accompanies all screen sizes, making it more enjoyable to users not on a mobile device

- This weeks focus for both Salmon Run and Solar Escape was on their accessibility. Alicia had informed us on the certain web guidelines that were essential in our game development
- It had given me a new lens that I had never thought of as deeply previously, throughout the coming weeks I would learn more about both web and video game accessibility and how vital it is to many users with disabilities and how simple these features can be added as long as it is not an afterthought but thought of in the initial planning. All my ideas and features from then on continue to get better and more refined with accessibility being something that comes first and not last.
- Solar Escape was lacking keyboard navigation of the ui and the gameplay, which was originally meant for only mouse and touch controls
- It was a major change to make the game more accessible as it was already complete, which taught me the importance of good planning of accessibility instead of considered after
- The learning process with godot was ongoing, using documentation and the help of others from the team to help guide me in the right direction of navigating and updating different components of the game in Godot. Videos online were very helpful in explaining basics that I had not yet learned and help improved the speed at which the accessibility changes were implemented if I hadnt asked those questions or did my own research.
- The work on these changes continued on to week 5 but it was a good start with getting my hands dirty in these programs that I had still been learning about, and it is great experience as I continue to take what i learn and apply it to future projects as well.

Week 5 (Feb 2-6)

- By week 5, there was a lot of talk about the new game possibilities for the upcoming launch of the website. Both ideas were from this team and the previous team were already put down in a spreadsheet to be then talked about for what would be the games we decide to move forward with. I had asked my supervisor Kierstan if I could share my idea with the rest off the digital team during the scrum, to both practice my presentation and communication skills, but to also provide my own input towards what is decided as the games we focus on for the rest of the coop term.
- The idea was a tower scaling game that would incorporate the science by separating the tower into different biomes, in a way to explain different creatures while keeping the gameplay exciting
- Initial feedback from the team was that they had liked the idea, and that it was interesting gameplay wise while still keeping the science relevant, however this game idea never then came to light as a new idea that I would later present would be something that could build on this original idea
- Us 4 coops had taken the time to update the spreadsheet of game ideas and add our own, from simulation style games to taking care of a creature game, to tower scaling games, we had built on the original list, doubling what was there before. We were all passionate about creating fun and interactive games that would help teach young users various STEM subjects in a way that they learn without knowing that they are learning

- Another idea of mine was to take the biome and different sciences approach and put it to a game hub open world region, that would let it be a project that is never completed as new science topics and regions can be built onto this world giving, creating an entrypoint for all the sciences that the user can then explore deeper by going to our other games and articles
- It gave me an opportunity again to present my idea to the team, building my own communication skills
- The digital team kept these ideas to the team, leaving next week where we would finalize and then share with both science and writing team
- I had completed my work with Solar Escape, the accessibility issues were resolved with a virtual pointer that would help you interact with objects in the scene, and universal buttons for accessing and navigating the pause menu and settings.
- Feedback from the team throughout the week on the changes help me resolve bugs and make changes that I had not previously thought of, especially in the accessibility sense. Making sure that buttons are not bound to keys that a user could accidentally press is an example of something that I had learned was important to think about, as well as the other guidelines for contrast in colours of objects and texts to make sure that items could be seen by everyone

Week 6 (Feb 9-13)

- This week marked finalization of the games list, and with all the ideas I had presented and shared with the team, my third idea on a pollution climate change based game is what ended up being one of the hero projects for the coop term. Learning from and getting feedback from my previous ideas help me build and improve from the ideas I had before to something that would make the biggest difference and impact to the users learning and how they think about the world today.
- Other interactives were discussed as part of the game library for the launch of the website including a space interactive that I was assigned to.
- The approach with the interactive had me shift away from the 2d game space to working with 3d models in three.js.
- This was an environment that I was also unfamiliar with, no one from the team had worked with this library before as all the project before stayed strictly 2d and 2d items. Both Joey and I had our interactives, his was based on human anatomy and mine was on space exploration. We had used what we learned and discussed constantly with one another on things that could improve each others work with features that we were still learning about.
- Both of our projects would have us working closely with the ui coops that were building up the 3d assets, it helped me learn from their side as well, shadowing their work to learn more about the certain art principles that come from both game design and interactives. I had wanted to learn about the various aspects of game development as one of my goals at this coop and this was a perfect opportunity for the continuing weeks of the rest of the coop to better learn from both the development and art side of how to make strong engaging interactives and games

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Week 7 (Feb 16-20)

- A lot of work in this week was working from the initial proof of concept and demo that I had shared with the team in our scrums of what is possible with 3d models in an interactive, taking what I had learned about in the library and the communication skills I had learned from sharing my previous ideas
- This week was focused on the science aspects of the building interactive, closely working and communicating with the science team on the ideas and approaches which I then later applied in the interactive
- It was a new experience for me to do more communication with the science team than the digital team. The science team were less familiar with games and digital concepts which had me learn how to communicate ideas with less technical terms that can be understood better to teams that are unfamiliar with programming concepts.
- The bulk of the workflow had me taking time to watch videos and read documentation that I had then later applied to the interactive, asking and answering Joe's questions from what we both were learning about the library
- By the end of the week we had a beta version of the interactive working as a scrollable with copy given from the science and writing team to make the first full interactive that I had worked on from start to finish

Week 8 (Feb 23-27)

- Week 8 commenced the start of the ideation phase of my hero piece idea
- I was excited to be working on a project that I had thought of that also gives me a chance to use it as a project to show what I have learned from this coop into a game that I can look back and be proud of
- Through the help of Kierstan and the Science lead Mary Jane, they had created the team of the writer, science lead, developer, and art designer that would be working on this long term project
- Our initial meeting wouldn't start until the week 9 but having this start was very exciting for what would come later in the coop
- Positive and important feedback and bugs were provided from the space interactive that I had been working on during the month. Not only the digital team, but the writing and science team were able to try and give feedback on the interactive, getting perspectives from teams with familiarity and experiences in different fields helped in creating a variety in the group that gave feedback, learning things about how some features of the interactive scrollable may be clear to the developer of the project but proper instructions and clear navigation are what I took to then help improve the experience for any user improving the overall quality of the interactive
- These changes were then seen with a positive outlook to everyone from the teams as they can better understand and enjoy and learn from the experience because of it

Week 9 (Mar 2-6)

- Week 9 marked the start of the hero piece game project that take till the end of the year leading up to the website launch
- It has been overall a great experience for learning and sharing with others for the past two months

- Everyone on the team was very helpful and communicative which helped me become more open with the team of my own ideas and feedback, helping me build confidence and communication skills that I can take outside of this coop.
- Going back to the godot engine after taking the break with the interactive, I had to relearn some concepts and revisit videos and documentation that I had previously explored to get a sense of the project space again
- Working from the very beginning of this game project versus near the end will provide me the best opportunity to take everything that I had learned about accessibility and educational interactives to build from the ground up keeping those different lenses in mind.
- With our kickoff the team had decided to take a science fiction approach with a strong underlying message of climate change and pollution that we hope will make a significant impact on the younger generation to continue the work and explore the sciences to make the changes for the better in our world
- It is always a learning experience and the open communication of the team is what promotes the productivity of myself and others to be inspired to do better and always improve

1. Learning Objective (This is what I want to learn)

I want to learn about the elements of game development using Godot as the game engine.

Rationale (This is why I want to learn it)

I want to learn more about it so that I can take those skills to continue making more games in the future.

Learning Activities (This is how I might learn it)

Work through a Godot game project

Follow tutorials online

Ask my team for help with the different functionalities of a game engine

2. Learning Objective (This is what I want to learn)

I want to improve my skills in presenting game ideas to different types of audiences with expertise in video games.

Rationale (This is why I want to learn it)

I want to improve this skill because it will help to receive feedback and support on different ideas that I come up with.

Learning Activities (This is how I might learn it)

Practice presenting the team with various game ideas and then receiving feedback on them.

Talk with the team about presenting strategies that help hook audiences less familiar with video games

Continuing to come up with new ideas and being open to sharing with the team or other departments

3. Learning Objective (This is what I want to learn)

I want to learn about the different aspects of visuals related to video games

Rationale (This is why I want to learn it)

I want to learn more about the visual UI/UX side because it would help me understand how those visuals will be applied across various aspects of a game.

Learning Activities (This is how I might learn it)

- Shadow work from the ui/ux team
- Ask questions about why certain choices were made on material and style
- Learn about the process behind creating visuals from sketches to sprites to scenes that come together to make the game

4. Learning Objective (This is what I want to learn)

I want to learn more about music composition, and how it plays a role in game development

Rationale (This is why I want to learn it)

I want to learn more about music composition as it has a key role in producing quality games. Understanding the concept will help me, as a developer, better understand its application and where it could make the biggest impact.

Learning Activities (This is how I might learn it)

- Ask members of the team about their experience with composing
- Learning music theory to train my ears on various sounds
- Practice creating short pieces and receiving feedback
- Apply the learning directly to projects that you are currently developing

5. Learning Objective (This is what I want to learn)

I want to learn more about the importance of accessibility in the visuals and mechanics of a game

Rationale (This is why I want to learn it)

I want to learn more about accessibility because it helps to better understand how we can reach the greatest audience and make the biggest impact with the games we make. Having a mindset that puts accessibility first so that the game that is made is enjoyable to all

Learning Activities (This is how I might learn it)

- Talk with the team about the guidelines for accessibility
- Explore the impacts of a game built with or without accessibility in mind
- Receive feedback on games created by audiences that require different accessibility to learn more about how we can be more inclusive

SMART Goals Evaluation

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Competency 1: Agility and Adaptability

Rationale:

In the rapidly changing world of technology and software, it is essential to demonstrate strong agility and adaptability. From my self-assessment, I had a strong desire to improve my ability to handle fluctuating workloads and learn new procedures required by my software engineering degree and co-op. By focusing on this competency, I aim to better handle uncertain, ambiguous, and unpredictable situations. Developing a structured approach to learning new tools, such as the Godot engine, and managing time effectively will ensure I can maintain high performance during heavy workload periods, such as exam seasons or project milestones.

S.M.A.R.T. Goal 1:

By the end of the next four months, I will master the core functionalities of the Godot game engine by completing one fully functional game prototype. Progress will be measured by the successful implementation of three key systems—player movement, UI, and save states—and a final review session with my team lead to evaluate technical proficiency.

S.M.A.R.T. Goal 2:

To improve my time management and workload handling, I will create a weekly calendar progress tracker. I will provide my team lead with a mid-week status report every Wednesday for the next six months, measuring success by my ability to meet 100% of project deadlines or communicate delays at least 48 hours in advance.

Competency 2: Effective Oral and Written Communication

Rationale:

Effective oral and written communication is crucial in determining how your ideas are perceived. While I currently feel comfortable in neutral communication settings, in the self-assessment I have identified an importance to improve my ability to "convince others that my ideas are valid" and "extract information from others". In a professional work environment, the ability to present technical game ideas clearly to various audiences is essential for receiving constructive feedback and securing project support. This competency will help me bridge the gap between creative vision and technical execution.

S.M.A.R.T. Goal 1:

Within the next six months, I will improve my professional presentation skills by volunteering to lead three 10-minute team summaries or project updates using PowerPoint. I will evaluate my growth using a feedback rubric provided to my audience to track specific improvements in clarity, delivery, and audience engagement.

S.M.A.R.T. Goal 2:

By the end of my 12-month co-op, I will improve my collaborative communication by conducting 12 monthly "Insight Sessions" with the UI/UX team. During each session, I will interview the team to document specific strategies they use for asset and environment building, measuring

success by applying at least one specific technique per month to my final project and summarizing these choices in a final reflection paper.

Competency 3: Curiosity and Imagination

Rationale:

Curiosity and imagination are what drive risk-taking to find innovative solutions. Through my self-assessment, I highlighted a strong interest in learning things that are new to me and playing with ideas beyond the first solution that comes to mind. In game development, this curiosity is best applied to areas like game mechanics and storytelling, through visuals and music composition. By focusing on this competency, I will ensure that the games I develop are inclusive and reach the widest possible audience while incorporating creative elements that make the biggest impact on user experience.

S.M.A.R.T. Goal 1:

By the end of my 12-month co-op, I will develop a "Mindset of Accessibility" by researching and applying Web Content Accessibility Guidelines (WCAG) to my final game project. Achievement will be measured by conducting at least three playtest sessions, both internally and with at least three users requiring different accessibility features, incorporating their feedback into the final build.

S.M.A.R.T. Goal 2:

To broaden my technical imagination, I will complete an introductory course in music theory and composition within the next eight months. I will measure this goal by composing and integrating three original short audio pieces into a Godot project and seeking feedback from team members with a music background on their application and impact on my final game project.

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